

D - Reindeer and Sleigh

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 400 points

Problem Statement

There are N sleighs numbered $1, 2, \dots, N$.

R_i reindeer are required to pull sleigh i .

Additionally, each reindeer can pull at most one sleigh. More precisely, $\sum_{k=1}^m R_{i_k}$ reindeer are required to pull m sleighs $i_1, i_2, \dots, i_m$.

Find the answer to Q queries of the following form:

- You are given an integer X . Determine the maximum number of sleighs that can be pulled when there are X reindeer.

Constraints

- $1 \leq N, Q \leq 2 \times 10^5$
- $1 \leq R_i \leq 10^9$
- $1 \leq X \leq 2 \times 10^{14}$
- All input values are integers.

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Input

The input is given from Standard Input in the following format:

```

 $N$   $Q$ 
 $R_1$   $R_2$   $\dots$   $R_N$ 
query1
query2
⋮
query $Q$ 

```

Each query is given in the following format:

```

 $X$ 

```

Output

Print Q lines.

The i -th line should contain the answer to the i -th query.

Sample Input 1

[Copy](#)

```

4 3
5 3 11 8
16
7
1000

```

[Copy](#)

Sample Output 1

[Copy](#)

```

3
1
4

```

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When there are 16 reindeer, sleighs 1, 2, 4 can be pulled.

It is impossible to pull four sleighs with 16 reindeer, so the answer to query 1 is 3.

Sample Input 2

[Copy](#)

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Sample Output 2

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Custom Test (/contests/abc334/custom_test)

Editorial (/contests/abc334/editorial)

1
2
2
3

Sample Input 3

Copy

```
2 2
1000000000 1000000000
2000000000000000
1
```

Copy

Sample Output 3

Copy

```
2
0
```

Copy

Language

C++23 (Clang 21.1.0) ▾

Source Code



Open File



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1 |

* at most 512 KiB



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